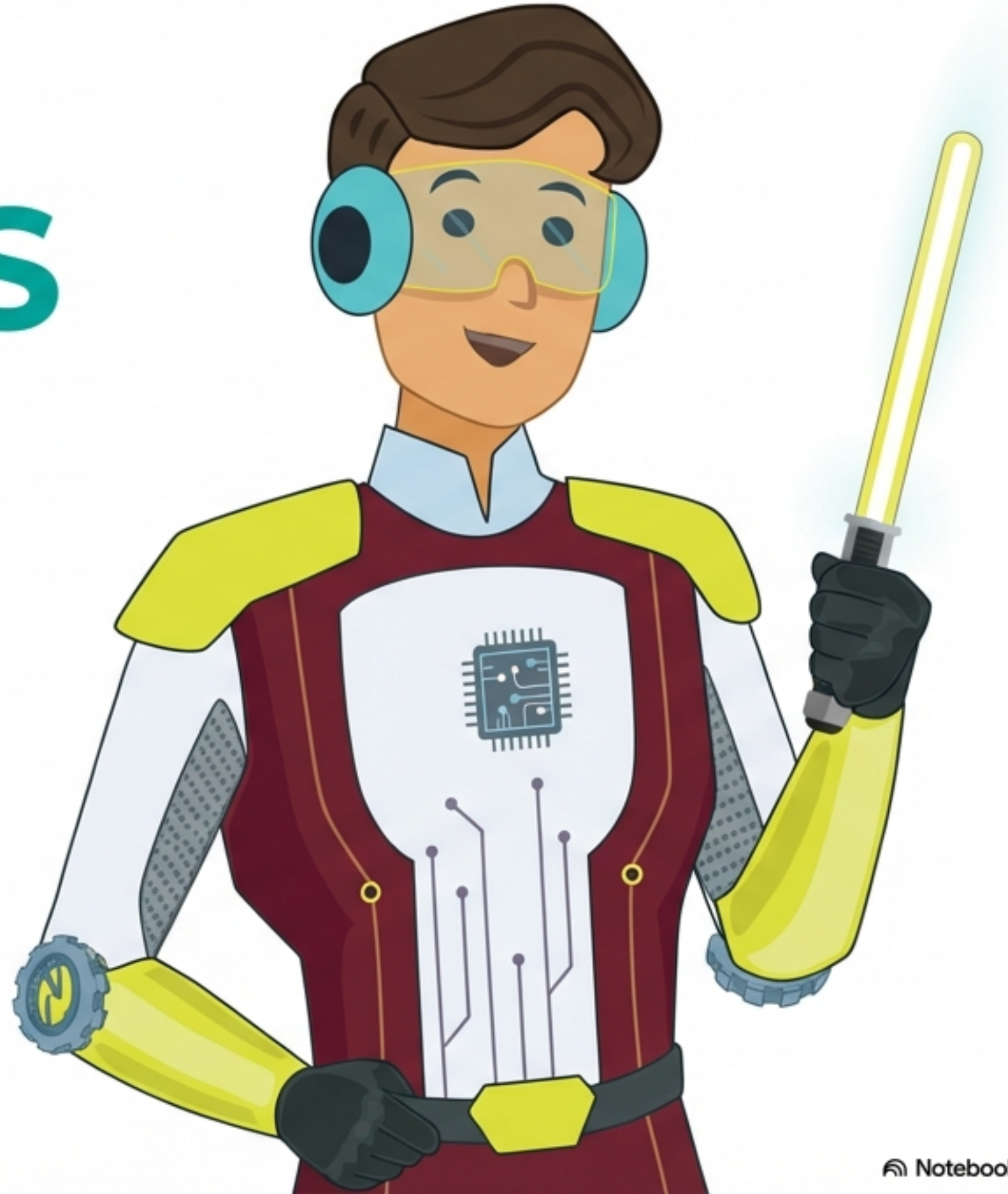


BASIC ELECTRONICS

Powering Our World

Grade 6 Engineering Workshop

Understanding the invisible power that lights our homes and runs our gadgets.



What is Electricity?

Electricity is a form of energy that flows through materials to help us do work.



Heating

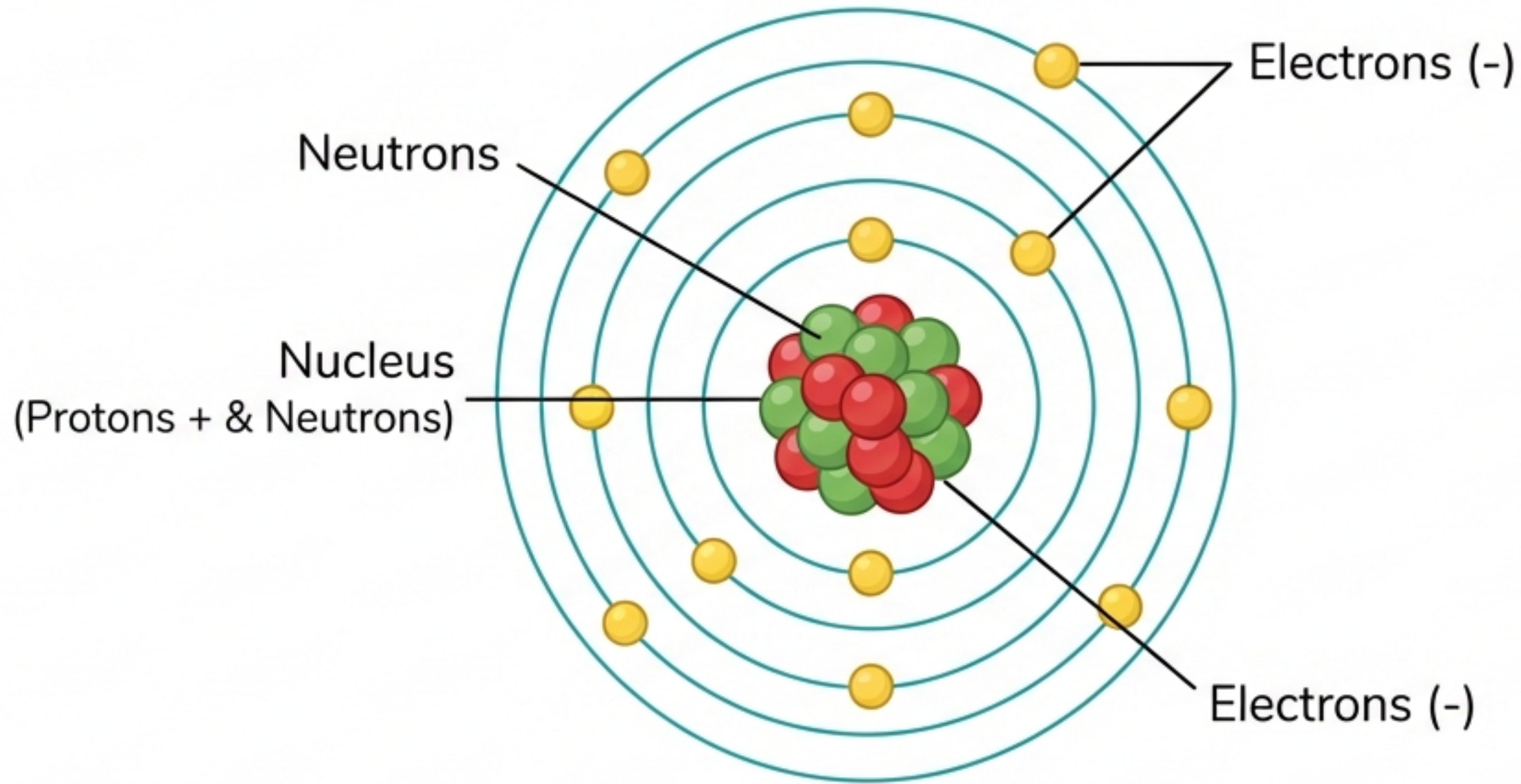


Lighting



**Powering
Machines**

The Building Blocks: Atoms



- **Protons:** Positive Charge (+)
- **Electrons:** Negative Charge (-)
- **Neutrons:** No Charge

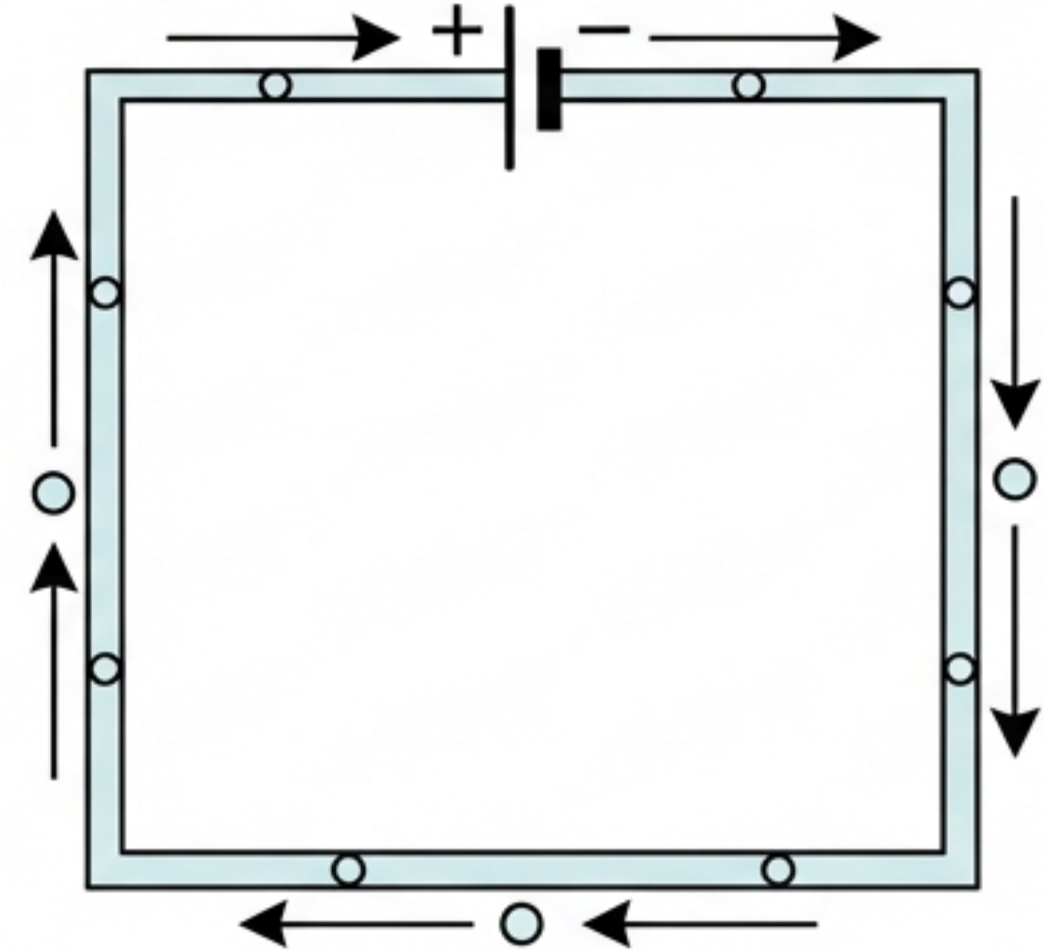
Key Insight

Opposites Attract! Positive protons pull on negative electrons.

How Current Flows



Water Current: Flows in a pipe



Electric Current: Electrons flow in a wire

Technical Rule: Current flows from Positive (+) to Negative (-).

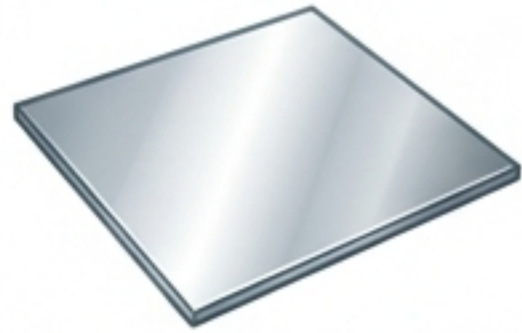
The Path: Conductors vs. Insulators

CONDUCTORS

Helpers that allow flow.



Copper wire



Aluminum sheet



Key



Coin

INSULATORS

Blockers that stop flow.



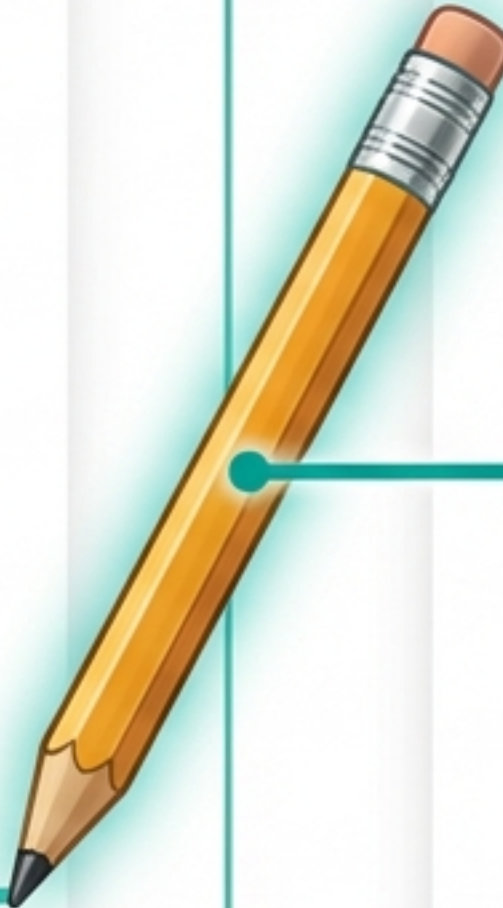
Log of wood



Rubber band



Plastic bottle



Where Does Energy Come From?

Renewable (Infinite)



Solar



Wind

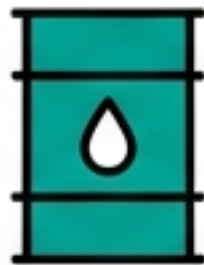


Hydro



Geothermal

Non-Renewable (Limited)



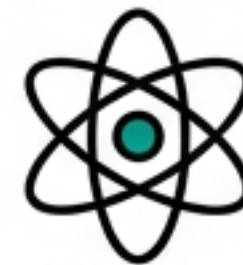
Fossil Fuel



Coal



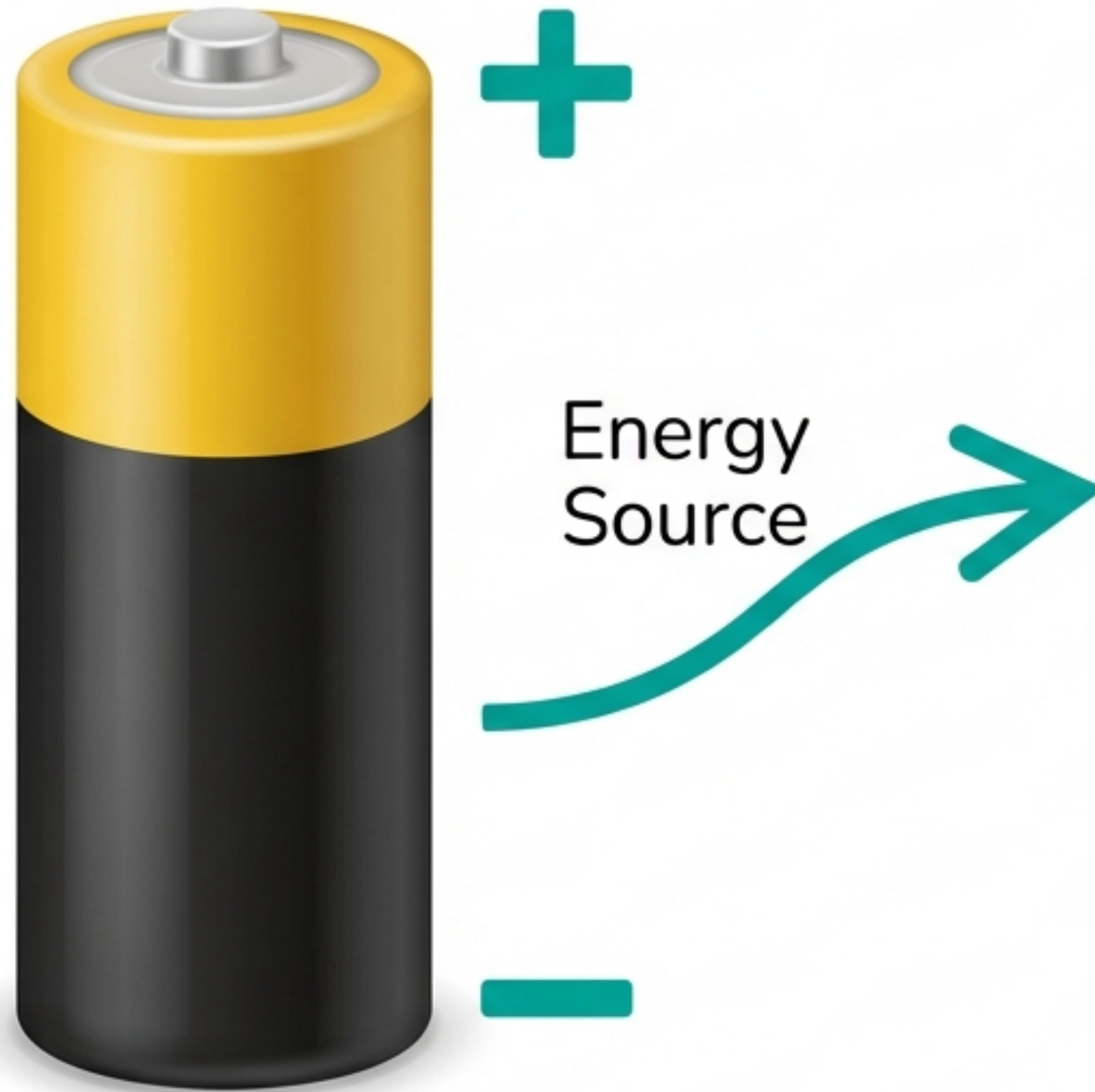
Natural Gas



Nuclear

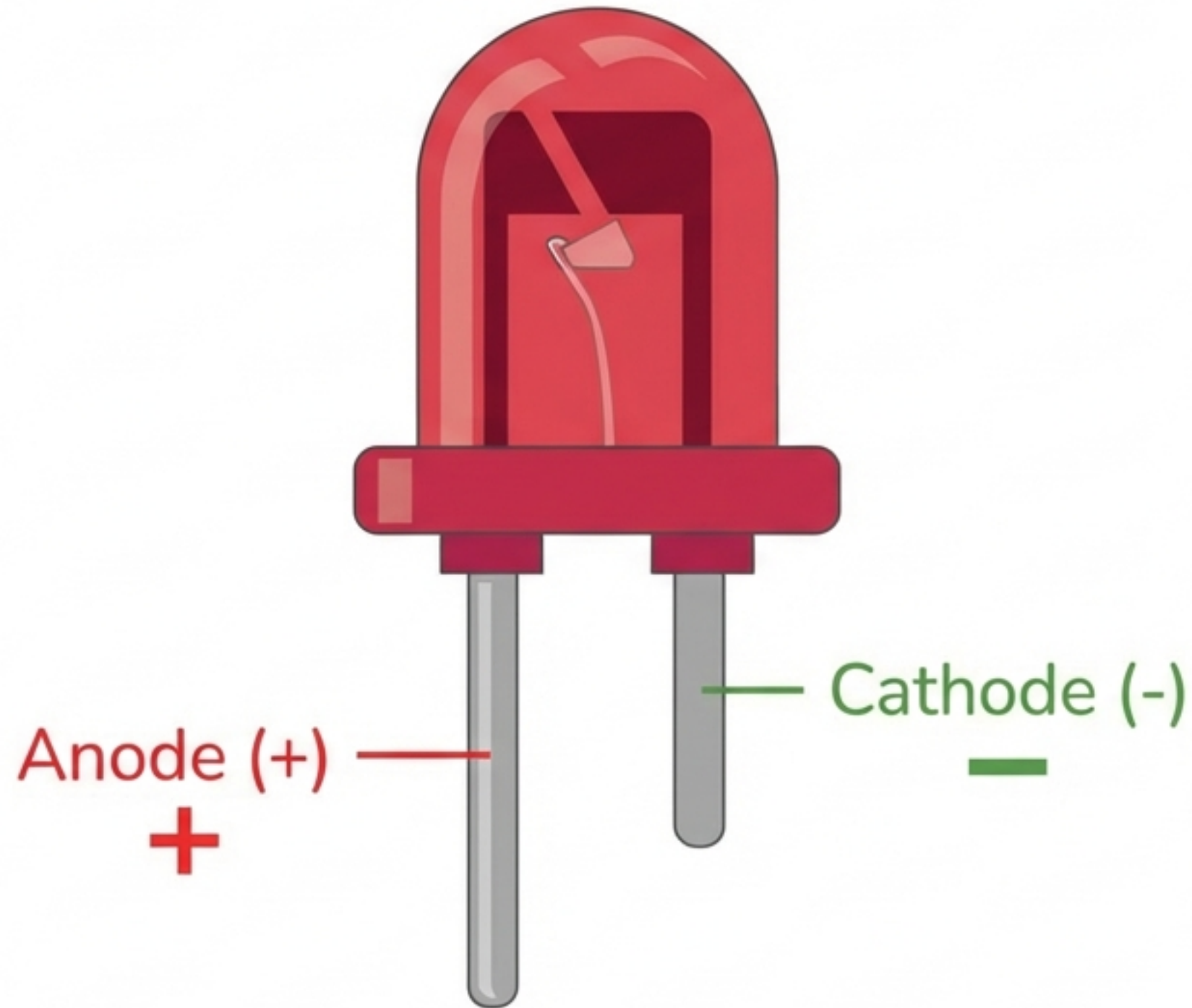
Energy cannot be created or destroyed. It can only be transformed.

The Battery: Energy Storage



- A container that stores chemical energy.
- Converts Chemical Energy into Electrical Energy.
- Pushes electrons through the circuit.

Meet the LED (Light Emitting Diode)



An electronic component that lights up when current flows through it.



Warning: Must connect **Positive to Positive!** If connected backwards, it won't light up.

Safety Protocol: Healthy Practices



NEVER put objects in outlets.



Keep electronics away from water.



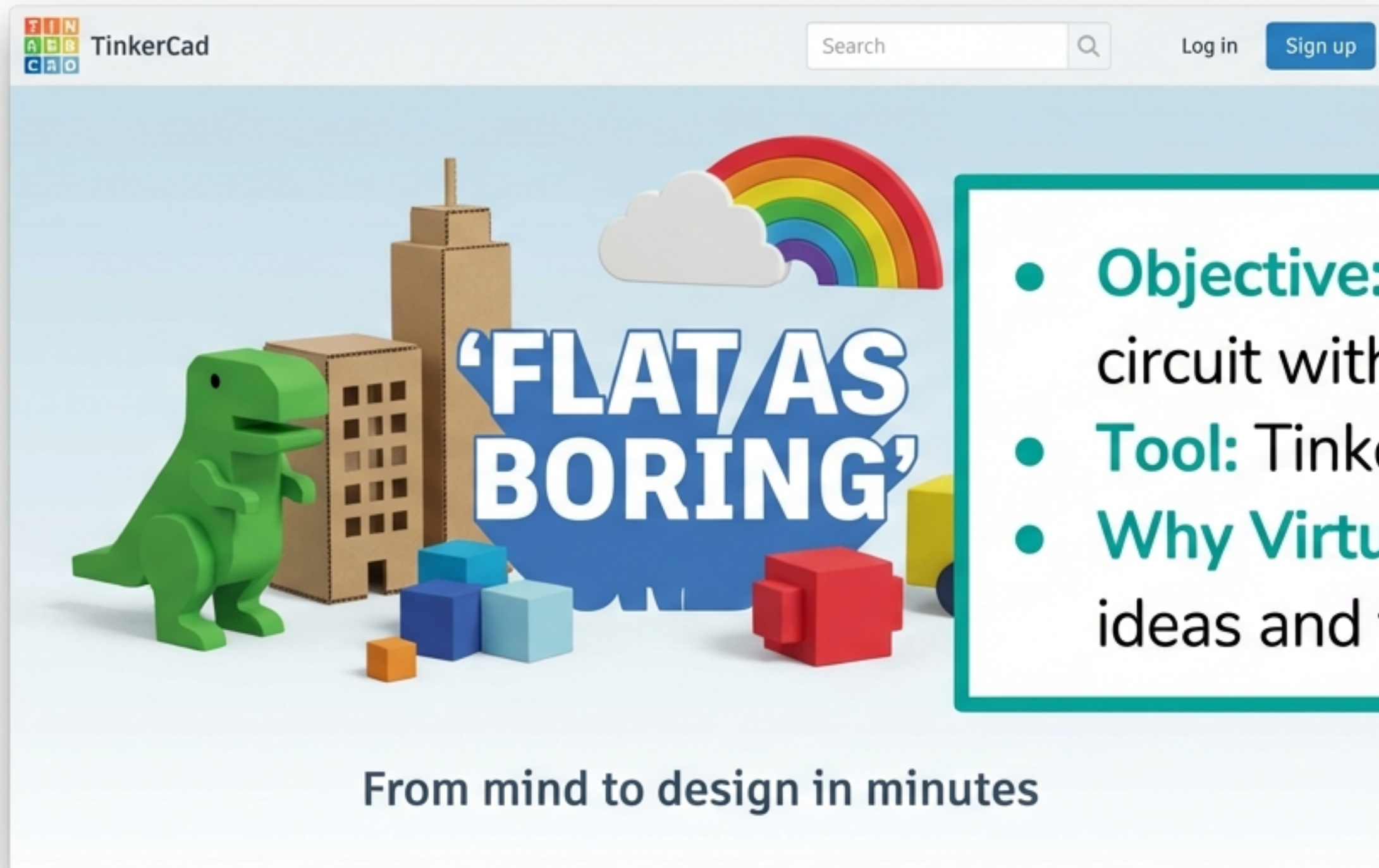
Do not touch switches with wet hands.



Keep cords away from heat.

Not sure? Ask an adult.

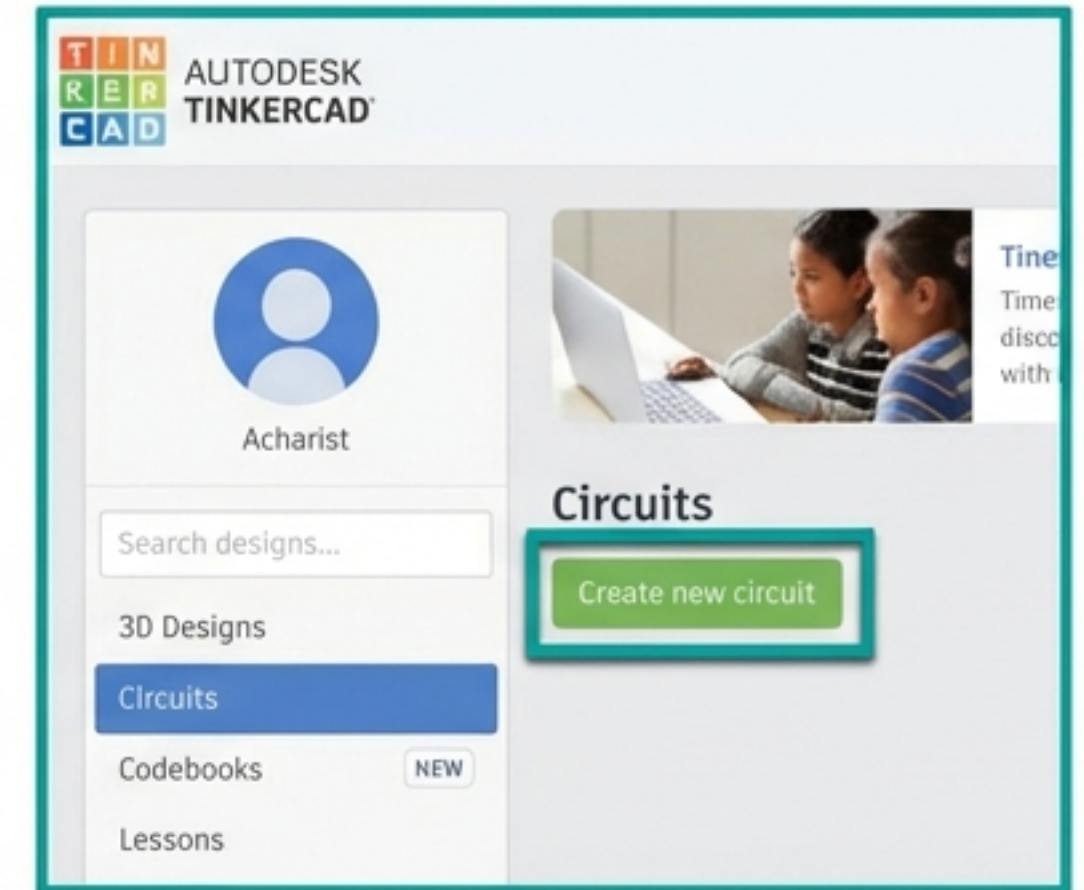
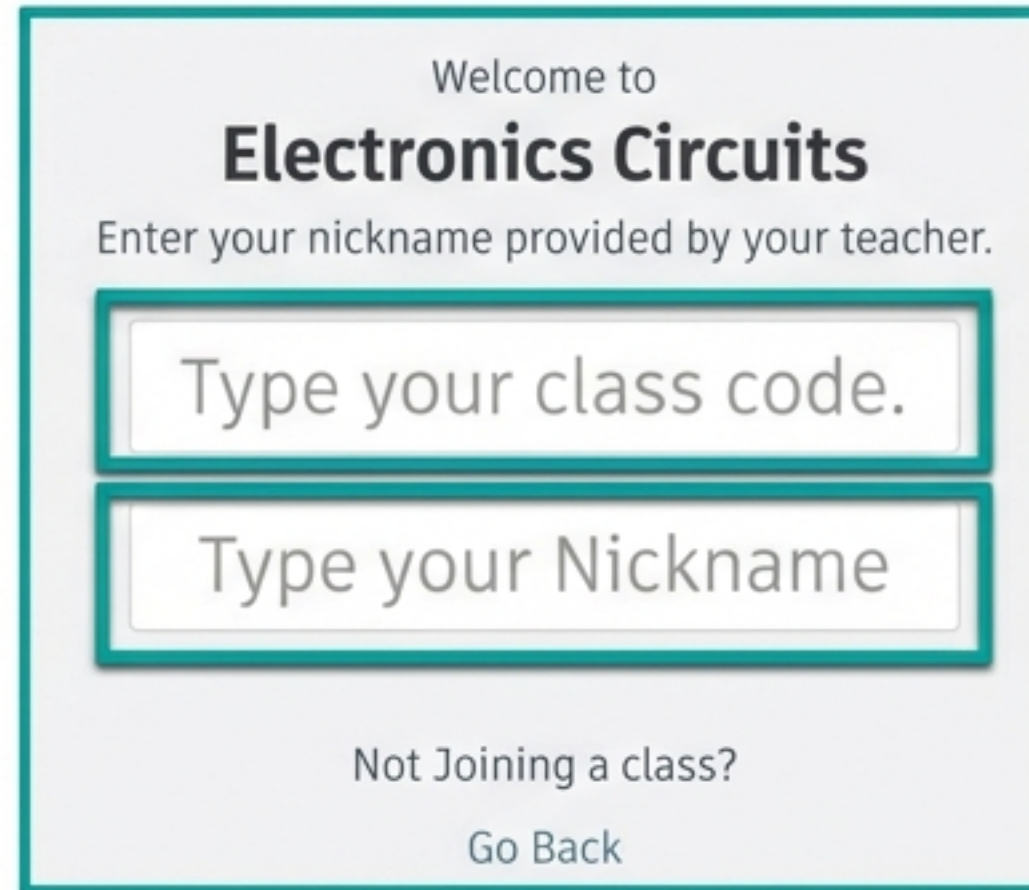
The Mission: Virtual Circuits



- **Objective:** Design a working circuit without getting shocked!
- **Tool:** Tinkercad.com
- **Why Virtual?** It allows us to test ideas and fix mistakes safely.

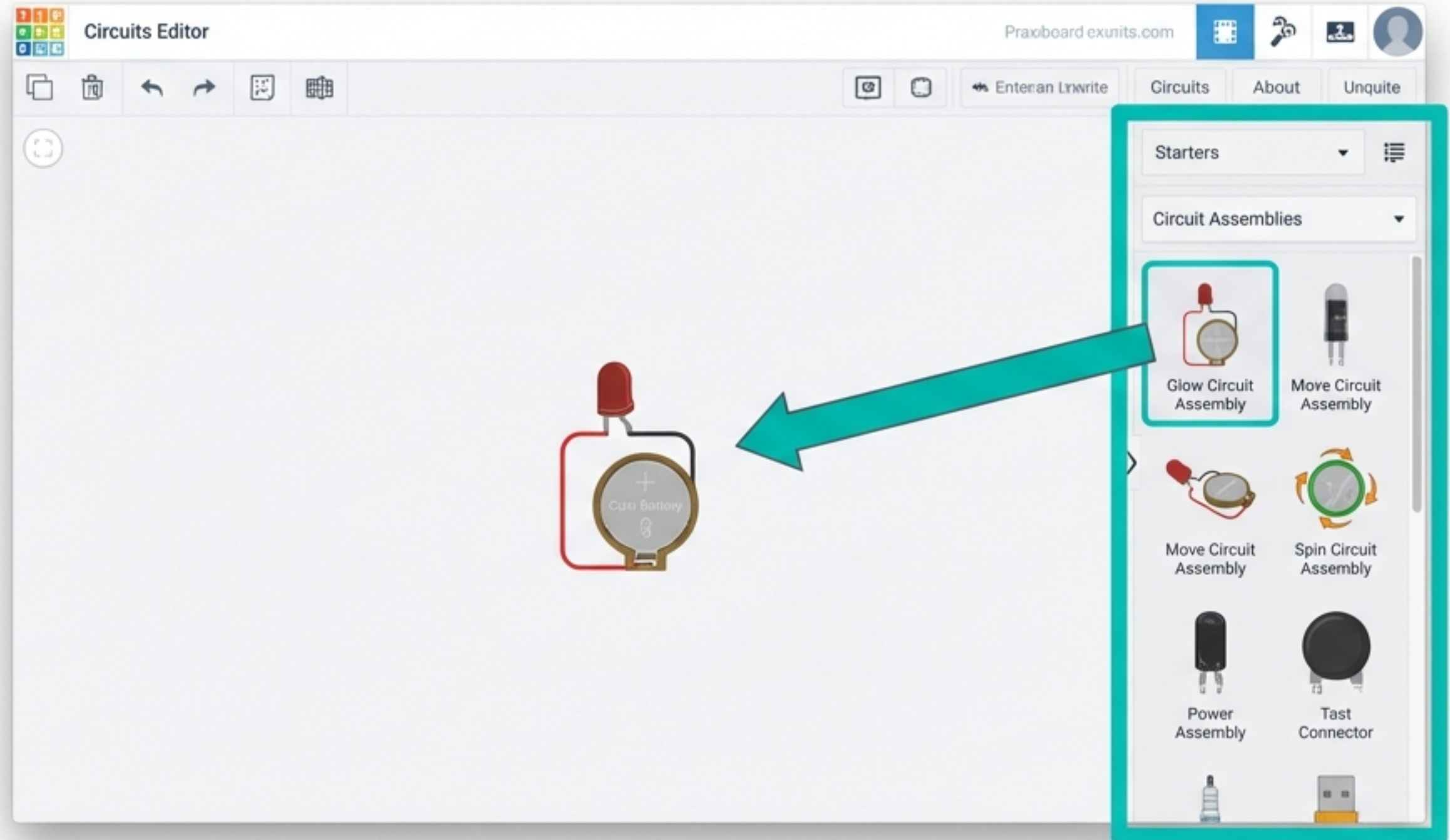
Mission Setup

- 1 Go to TinkerCad.com and click Join Class.
- 2 Enter Class Code & Nickname.
- 3 Click Create New Circuit.

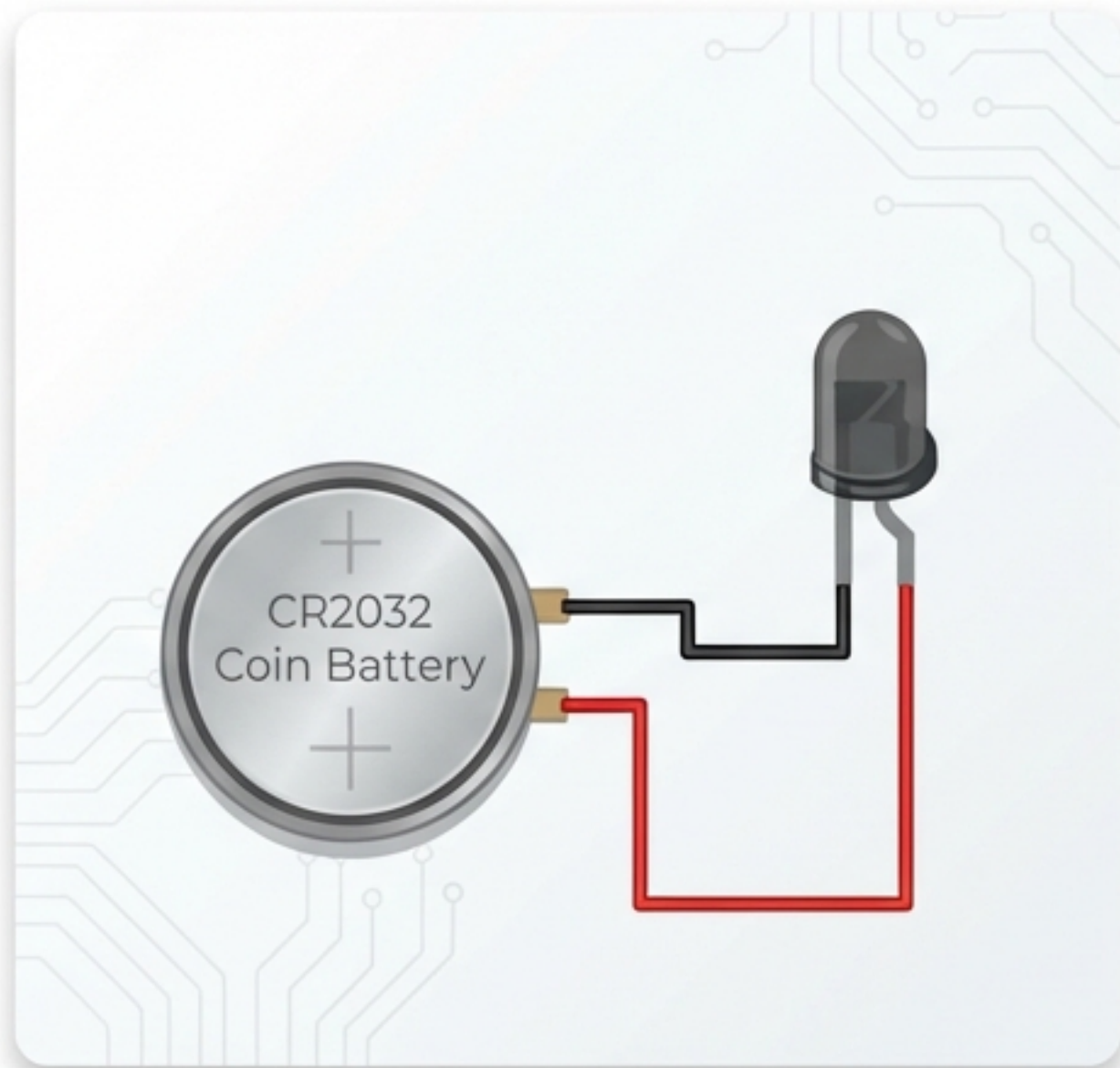


Building the Glow Circuit

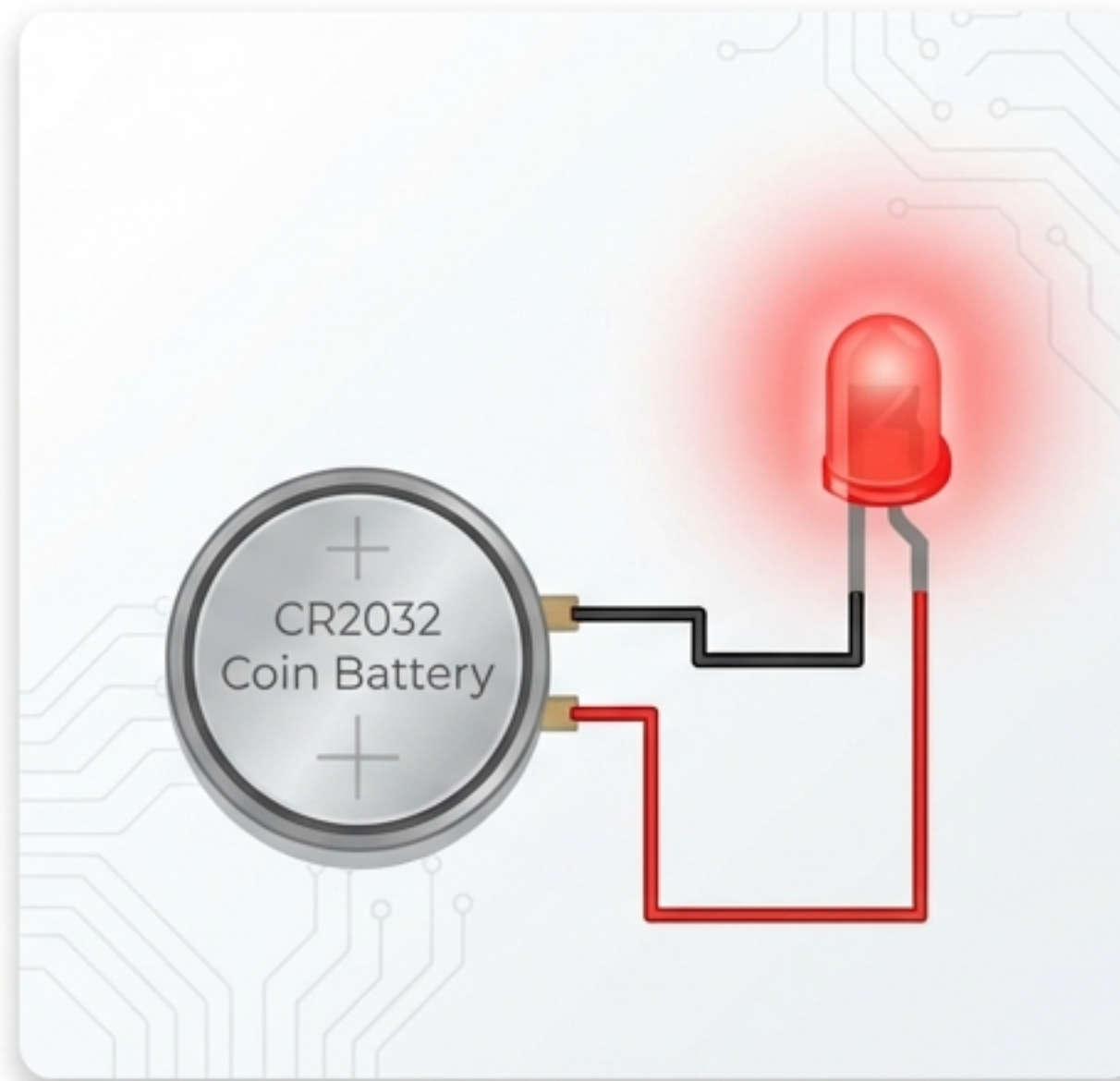
Find 'Circuit Assemblies' in the Starters menu. Drag the Glow Circuit to the workspace.



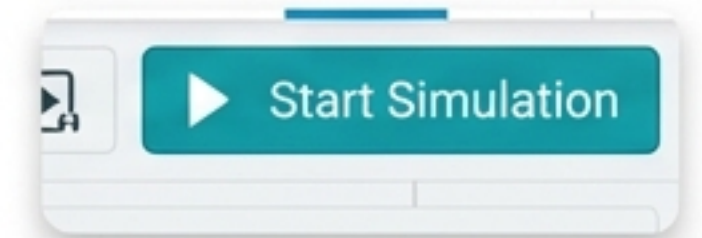
Simulation Time!



Simulation Stopped



Simulation Running



Click **Start Simulation**.

Is your LED
glowing red?
Success!

Engineer Knowledge Check

Question 1

True or False?
Current flows
Negative to Positive.



Answer:
FALSE (It flows
Positive to
Negative).

Question 2

True or False?
Coal is a non-
renewable resource.



Answer:
TRUE.

Question 3

Which LED leg is
Positive?



Answer:
The **Long Leg**
(**Anode**).

Tech Flash: Mission Complete

Vocabulary Recap

- **Electricity:** Energy carried by wires.
- **Conductor:** Allows flow (Copper).
- **Insulator:** Stops flow (Rubber).
- **Current:** The flow of electrons.

Great job, Future Engineers!

